

A Superfast Direct Solver for 2D Type-II Inverse Nonuniform Discrete Fourier Transform Based on Hierarchically Semiseparable Matrices

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September 4, 2026

Joint work with Prof. Yingzhou Li

- 1 Introduction
- 2 Review of the 1D Method
- 3 Low-Rank Property of the NUDFT Matrix
- 4 Approximate Direct Inversion of the 2D Type-II NUDFT Matrix
- 5 Numerical Examples

Two-dimensional **nonuniform discrete Fourier transform (NUDFT)**:

$$f_j = \sum_{k^{[x]}=0}^{n^{[x]}-1} \sum_{k^{[y]}=0}^{n^{[y]}-1} c_{k^{[x]}, k^{[y]}} e^{-2\pi i(k^{[x]}x_j + k^{[y]}y_j)}, \quad 0 \leq j \leq M-1.$$

- Sample points: $(x_j, y_j) \in [0, 1)^2$;
- Frequencies: $(k^{[x]}, k^{[y]}) \in \{0, 1, \dots, n^{[x]} - 1\} \times \{0, 1, \dots, n^{[y]} - 1\}$;
- Coefficients: $c_{k^{[x]}, k^{[y]}} \in \mathbb{C}$;
- Target values: $f_j \in \mathbb{C}$;
- Number of sample points M , number of frequencies $N = n^{[x]}n^{[y]}$.

NUDFT matrix:

$$\mathbf{A} = \left(e^{-2\pi i(k^{[x]}x_j + k^{[y]}y_j)} \right)_{j, (k^{[x]}, k^{[y]})} \in \mathbb{C}^{M \times N}.$$

Forward and Inverse Problems

Forward problem: $\mathbf{f} = \mathbf{A}\mathbf{c}$.

- NUFFT [Barnett, Magland, and af Klinteberg 2019]
Complexity: $\mathcal{O}(M + N + N \log N)$.

Inverse problem: $\min_{\mathbf{c} \in \mathbb{C}^N} \|\mathbf{A}\mathbf{c} - \mathbf{f}\|_2, \mathbf{c} = \mathbf{A}^\dagger \mathbf{f}$.

- Iterative methods: LSQR + NUFFT;
- Direct methods: [Kircheis and Potts 2023].

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Main Idea

1D NUDFT matrix:

$$\mathbf{A} = \left(e^{-2\pi i k x_j} \right)_{j,k} \in \mathbb{C}^{M \times N} \quad \text{with} \quad x_j \in [0, 1).$$

1D DFT matrix:

$$\mathbf{F} = \left(\mathbf{F}(k, \ell) = e^{-2\pi i k \ell / N} \right)_{k,\ell} \in \mathbb{C}^{N \times N}.$$

The matrix $\mathbf{G} = \mathbf{A}\mathbf{F}^{-1}$ can be expressed as a **kernel matrix**:

$$\mathbf{G}(j, k) = \frac{1}{N} \frac{(\gamma_j / \xi_k)^N - 1}{(\gamma_j / \xi_k) - 1} =: \Psi(\gamma_j, \xi_k).$$

- $\gamma_j = e^{-2\pi i x_j} \in \mathbb{S} := \{z \in \mathbb{C} : |z| = 1\};$
- $\xi_k = e^{-2\pi i k / N} \in \mathbb{S}.$

$\rightsquigarrow$ Approximate $\mathbf{G}$ by a **hierarchically semiseparable (HSS) matrix**.

$\rightsquigarrow$ Use an HSS least-squares solver based on a URV factorization [Xi et al. 2014].

Low-Rank Property of the Transformed NUDFT Matrix

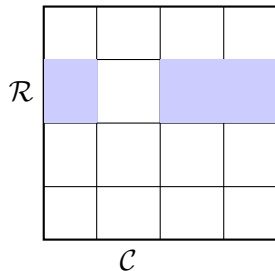
Theorem (Numerical rank of HSS blocks)

Let $[a, b) \subset [0, 1)$ be an interval, and define

$$\mathcal{R} = \{j : x_j \in [a, b)\} \subset \mathcal{J}, \quad \mathcal{C} = \{k : k/N \in [a, b)\} \subset \mathcal{K}.$$

Then, for $0 < \varepsilon < 1$, the HSS row block $\mathbf{G}(\mathcal{R}, \mathcal{C}^c)$ has ε -rank at most $\mathcal{O}(\log(1/\varepsilon) \log |\mathcal{C}|)$.

Wilber, Epperly, and Barnett 2025



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Face-Splitting of the NUDFT Matrix

NUDFT matrices in the x - and y -directions:

$$\mathbf{A}^{[x]} = \left(e^{-2\pi i k^{[x]} x_j} \right)_{j, k^{[x]}} \in \mathbb{C}^{M \times n^{[x]}} \quad \text{and} \quad \mathbf{A}^{[y]} = \left(e^{-2\pi i k^{[y]} y_j} \right)_{j, k^{[y]}} \in \mathbb{C}^{M \times n^{[y]}}.$$

Then

$$\mathbf{A} = \mathbf{A}^{[x]} \bullet \mathbf{A}^{[y]} = \begin{bmatrix} \mathbf{A}^{[x]}(0, :) \otimes \mathbf{A}^{[y]}(0, :) \\ \mathbf{A}^{[x]}(1, :) \otimes \mathbf{A}^{[y]}(1, :) \\ \vdots \\ \mathbf{A}^{[x]}(M-1, :) \otimes \mathbf{A}^{[y]}(M-1, :) \end{bmatrix} \in \mathbb{C}^{M \times N}.$$

Transformed NUDFT Matrix

DFT matrices in the x - and y -directions:

$$\mathbf{F}^{[x]} \in \mathbb{C}^{n^{[x]} \times n^{[x]}} \quad \text{and} \quad \mathbf{F}^{[y]} \in \mathbb{C}^{n^{[y]} \times n^{[y]}}.$$

Let $\mathbf{F} = \mathbf{F}^{[x]} \otimes \mathbf{F}^{[y]} \in \mathbb{C}^{N \times N}$ be the 2D DFT matrix. Then

$$\mathbf{G} = \mathbf{A}\mathbf{F}^{-1} = (\mathbf{A}^{[x]} \mathbf{F}^{[x],-1}) \bullet (\mathbf{A}^{[y]} \mathbf{F}^{[y],-1}) = \mathbf{G}^{[x]} \bullet \mathbf{G}^{[y]},$$

whose entries can be expressed as

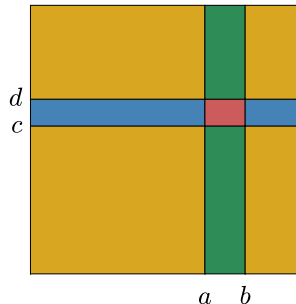
$$\mathbf{G}(j, (k^{[x]}, k^{[y]})) = \Psi^{[x]}(\gamma_j^{[x]}, \xi_{k^{[x]}}^{[x]}) \Psi^{[y]}(\gamma_j^{[y]}, \xi_{k^{[y]}}^{[y]}) = \Psi(\gamma_j, \boldsymbol{\xi}_{(k^{[x]}, k^{[y]})}).$$

- $\gamma_j^{[x]} = e^{-2\pi i x_j}$, $\gamma_j^{[y]} = e^{-2\pi i y_j}$, $\gamma_j = (\gamma_j^{[x]}, \gamma_j^{[y]}) \in \mathbb{S}^2$;
- $\xi_{k^{[x]}}^{[x]} = e^{-2\pi i k^{[x]}/n^{[x]}}$, $\xi_{k^{[y]}}^{[y]} = e^{-2\pi i k^{[y]}/n^{[y]}}$, $\boldsymbol{\xi}_{(k^{[x]}, k^{[y]})} = (\xi_{k^{[x]}}^{[x]}, \xi_{k^{[y]}}^{[y]}) \in \mathbb{S}^2$.

Intuition for the Low-Rank Property of the Transformed NUDFT Matrix

Submatrix of $\mathbf{G}$:

- $\mathcal{R}$: Row points in $[a, b) \times [c, d)$;
- $\mathcal{C} = \mathcal{C}^{[x]} \times \mathcal{C}^{[y]}$: Column points in $[a, b) \times [c, d)$;
- $\mathcal{C}^c = \mathcal{K} \setminus \mathcal{C} = \mathcal{C}_b \cup \mathcal{C}_g \cup \mathcal{C}_y$.



Intuition for the Low-Rank Property of the Transformed NUDFT Matrix

The submatrix can be partitioned as:

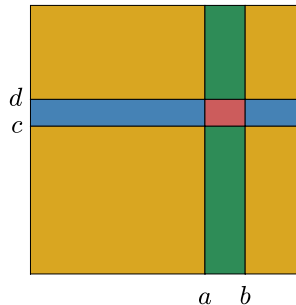
$$\mathbf{G}(\mathcal{R}, \mathcal{C}^c) = [\mathbf{G}(\mathcal{R}, \mathcal{C}_b) \quad \mathbf{G}(\mathcal{R}, \mathcal{C}_g) \quad \mathbf{G}(\mathcal{R}, \mathcal{C}_y)],$$

where

$$\mathbf{G}(\mathcal{R}, \mathcal{C}_b) = \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x],c})}_{\text{low-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y]})}_{\text{potentially full-rank}},$$

$$\mathbf{G}(\mathcal{R}, \mathcal{C}_g) = \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x]})}_{\text{potentially full-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y],c})}_{\text{low-rank}},$$

$$\mathbf{G}(\mathcal{R}, \mathcal{C}_y) = \underbrace{\mathbf{G}^{[x]}(\mathcal{R}, \mathcal{C}^{[x],c})}_{\text{low-rank}} \bullet \underbrace{\mathbf{G}^{[y]}(\mathcal{R}, \mathcal{C}^{[y],c})}_{\text{low-rank}}.$$



Intuition for the Low-Rank Property of the Transformed NUDFT Matrix

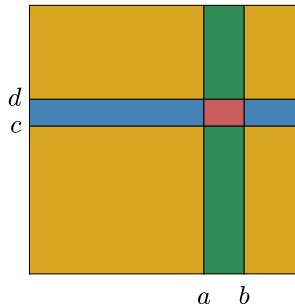
Lemma: $\text{rank}(\mathbf{A}_1 \bullet \mathbf{A}_2) \leq \text{rank}(\mathbf{A}_1) \text{rank}(\mathbf{A}_2)$.

Therefore, intuitively,

$$\text{rank}_\varepsilon(\mathbf{G}(\mathcal{R}, \mathcal{C}^c)) \lesssim \log(1/\varepsilon) n_c \log n_c,$$

where $n_c = \max\left\{\left|\mathcal{C}^{[x]}\right|, \left|\mathcal{C}^{[y]}\right|\right\}$.

$\leadsto \mathbf{G}$ admits an HSS approximation with HSS rank $\mathcal{O}(\log(1/\varepsilon)\sqrt{N} \log N)$.



Face-Splitting Product of HSS Trees

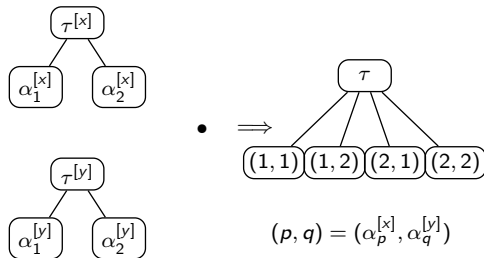
Suppose that $T^{[x]}$ and $T^{[y]}$ are full binary HSS trees of the same maximum level.

For $\tau = (\tau^{[x]}, \tau^{[y]})$, define

$$\text{ch}(\tau) = \text{ch}(\tau^{[x]}) \times \text{ch}(\tau^{[y]}).$$

The index sets are

$$\mathcal{J}_\tau = \mathcal{J}_{\tau^{[x]}} \cap \mathcal{J}_{\tau^{[y]}} \quad \text{and} \quad \mathcal{K}_\tau = \mathcal{K}_{\tau^{[x]}} \times \mathcal{K}_{\tau^{[y]}}.$$



binary $\bullet$ binary $\longrightarrow$ quadtree.

Closure under the Face-Splitting Product

Theorem (Li and L., 2026)

The face-splitting product preserves the HSS structure:

$\mathbf{T} = \mathbf{T}^{[x]} \bullet \mathbf{T}^{[y]}$ *is an HSS tree,*

$\mathbf{H} = \mathbf{H}^{[x]} \bullet \mathbf{H}^{[y]}$ *is an HSS matrix with respect to $\mathbf{T}$.*

Levelwise rank bound

Suppose that, at level ℓ ,

- the HSS ranks of $\mathbf{H}^{[x]}$ and $\mathbf{H}^{[y]}$ are bounded by $r_\ell^{[x]}$ and $r_\ell^{[y]}$;
- their maximum diagonal-block sizes are $d_\ell^{[x]}$ and $d_\ell^{[y]}$.

Then the HSS rank of $\mathbf{H}$ at level ℓ is bounded by $r_\ell^{[x]} d_\ell^{[y]} + r_\ell^{[y]} d_\ell^{[x]} + r_\ell^{[x]} r_\ell^{[y]}$.

Theorem (Li and L., 2026)

Suppose that

$$\|\tilde{\mathbf{G}}^{[x]} - \mathbf{G}^{[x]}\|_{\mathbf{F}} \leq \varepsilon \|\mathbf{G}^{[x]}\|_{\mathbf{F}} \quad \text{and} \quad \|\tilde{\mathbf{G}}^{[y]} - \mathbf{G}^{[y]}\|_{\mathbf{F}} \leq \varepsilon \|\mathbf{G}^{[y]}\|_{\mathbf{F}}.$$

Under a mild assumption, the face-splitting product $\tilde{\mathbf{G}} = \tilde{\mathbf{G}}^{[x]} \bullet \tilde{\mathbf{G}}^{[y]}$ satisfies

$$\|\tilde{\mathbf{G}} - \mathbf{G}\|_{\mathbf{F}} \leq (1 + C)\varepsilon \|\mathbf{G}\|_{\mathbf{F}}.$$

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The Construction of the HSS Approximation

Lemma (Interpolative decomposition, ID, Cheng et al. 2005): For a rank r matrix $\mathbf{A} \in \mathbb{C}^{m \times n}$, there exists a row subset $\hat{\mathcal{J}}$ of size r and an interpolation matrix $\mathbf{L} \in \mathbb{C}^{m \times r}$ such that $\mathbf{A} = \mathbf{L}\mathbf{A}(\hat{\mathcal{J}}, :)$.

To construct the HSS approximation of $\mathbf{G}$, we need to compute the row IDs of the HSS row blocks $\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c)$ for each node τ in the HSS tree:

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) = \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \mathbf{L}_\tau \Psi(\mathcal{Z}_{\hat{\mathcal{J}}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}),$$

where $\mathcal{Z}_{\mathcal{J}_\tau} = \{\gamma_j\}_{j \in \mathcal{J}_\tau}$ and $\mathcal{W}_{\mathcal{K}_\tau^c} = \{\xi_{(k^{[x]}, k^{[y]})}\}_{(k^{[x]}, k^{[y]}) \in \mathcal{K}_\tau^c}$.

However, directly computing the row ID of $\Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c})$ is expensive because $|\mathcal{K}_\tau^c|$ is large.

The Proxy Technique

For each node τ , let $\mathcal{P}_\tau$ be a set of **proxy points** such that

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) = \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \mathbf{Q}_\tau^*$$

for some matrix $\mathbf{Q}_\tau$.

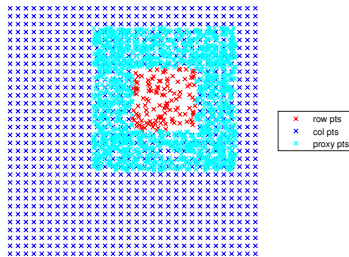
Then, if we compute the row ID

$$\Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{P}_\tau) \approx \mathbf{L}_\tau \Psi(\mathcal{Z}_{\hat{\mathcal{J}}_\tau}, \mathcal{P}_\tau),$$

it follows that

$$\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c) \approx \Psi(\mathcal{Z}_{\mathcal{J}_\tau}, \mathcal{W}_{\mathcal{K}_\tau^c}) \approx \mathbf{L}_\tau \Psi(\mathcal{Z}_{\hat{\mathcal{J}}_\tau}, \mathcal{P}_\tau) \mathbf{Q}_\tau^* \approx \mathbf{L}_\tau \mathbf{G}(\hat{\mathcal{J}}_\tau, \mathcal{K}_\tau^c).$$

Thus, the same interpolation matrix yields a row ID of $\mathbf{G}(\mathcal{J}_\tau, \mathcal{K}_\tau^c)$.



A Direct Solver for the 2D Type-II Inverse NUDFT Problem

Algebraic structure:

$$\mathbf{A} = \mathbf{GF} \approx \mathbf{A}_{\text{fast}} := \tilde{\mathbf{G}}\mathbf{F},$$

Algorithm A direct solver for the 2D type-II inverse NUDFT problem

Input: Sample points $\{(x_j, y_j)\}_{j=0}^{M-1}$, frequency counts $n^{[x]}$ and $n^{[y]}$, target values $\{f_j\}_{j=0}^{M-1}$.

Output: Coefficients $\mathbf{c}$.

- 1: Construct the HSS approximation $\tilde{\mathbf{G}}$ of $\mathbf{G} = \mathbf{AF}^{-1}$ using the kernel matrix expression.
 - 2: Compute the URV factorization of $\tilde{\mathbf{G}}$.
 - 3: Compute $\mathbf{c} = \mathbf{F}^{-1}\tilde{\mathbf{G}}^\dagger \mathbf{f}$ using the URV solve and a two-dimensional iFFT.
-

Complexities:

- **Offline stage** (construction and factorization): $\mathcal{O}(M + N^{3/2} \log^3 N)$;
- **Online stage** (HSS solve + iFFT): $\mathcal{O}(M + N \log^3 N)$.

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Time Scaling

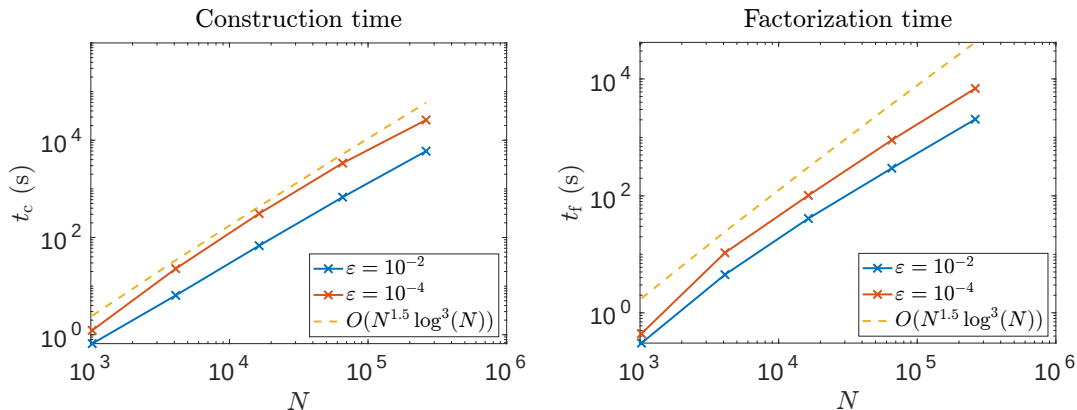


Figure: Construction time and factorization time of the direct solver on a polar grid.

Both times are consistent with the predicted $\mathcal{O}(N^{3/2} \log^3 N)$ scaling.

Iterative Solver

N	M	method	t_{pre}	t_{iter}	n_{iter}	r_s	e_s
64^2	11620	lsqr	-	2.9e+00	500	3.8e-03	4.0e-03
		plsqr ($\varepsilon = 10^{-2}$)	1.1e+01	2.2e+00	21	1.5e-11	1.6e-11
		plsqr ($\varepsilon = 10^{-4}$)	3.3e+01	1.4e+00	5	8.9e-11	9.3e-11
128^2	54373	lsqr	-	5.6e+00	500	2.2e-05	1.6e-03
		plsqr ($\varepsilon = 10^{-2}$)	1.1e+02	2.8e+01	38	2.6e-13	1.4e-11
		plsqr ($\varepsilon = 10^{-4}$)	4.1e+02	1.5e+01	10	2.2e-13	6.7e-11
256^2	249074	lsqr	-	1.7e+01	500	1.9e-03	1.8e-03
		plsqr ($\varepsilon = 10^{-2}$)	9.7e+02	1.3e+02	37	1.3e-11	1.2e-11
		plsqr ($\varepsilon = 10^{-4}$)	4.3e+03	1.0e+02	12	5.7e-11	5.5e-11
512^2	1122019	lsqr	-	6.5e+01	500	4.8e-04	4.6e-04
		plsqr ($\varepsilon = 10^{-2}$)	8.0e+03	4.2e+02	24	2.2e-12	2.1e-12
		plsqr ($\varepsilon = 10^{-4}$)	3.3e+04	4.2e+02	10	2.5e-12	2.4e-12

lsqr reaches the 500-iteration limit, whereas plsqr converges in **fewer than 40 iterations**.

- **Decomposition:**

$$\mathbf{A} = \mathbf{GF} \approx \tilde{\mathbf{G}}\mathbf{F}$$

$\mathbf{G}$: kernel matrix; $\mathbf{F}$: 2D DFT matrix; $\tilde{\mathbf{G}}$: HSS approximation.

- **Direct solver**

Offline complexity $\mathcal{O}(M + N^{3/2} \log^3 N)$ and online complexity $\mathcal{O}(M + N \log^3 N)$;

- **Effective preconditioner:**

On polar grids, the iteration count drops from the maximum of 500 to **fewer than 40**.

Thank you!

My homepage:

<https://jingyuliumath.github.io/>

Related paper:

<https://arxiv.org/abs/2607.00928>

